From Farm and Weather Data to Solar and Battery Trading Decisions

Data, ensemble formulation and real-time delivery

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Abstract

A solar producer must commit electricity before its production is known. Battery storage provides flexibility, while successive auctions create opportunities to revise the contracted position as weather information improves. We study this problem using four northern-Italian farms, historical production and irradiance, regularly updated weather forecasts, and 50-member weather ensembles. Weather is converted into possible AC production paths that are used to plan physically feasible trading and battery decisions. Auction participation studies investigate the importance of jointly optimizing current trades and future adjustments. This document explains the available data, the forecast-to-decision process, the deterministic and ensemble formulations, and physical delivery control. The full optimization and supporting equations are collected in appendices.

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1 The problem and the core idea

A photovoltaic (PV) farm and a battery share a grid connection. Solar generation can be exported, stored or curtailed; the battery can be charged by the local farm or by grid imports while also being able to discharge to the network. There is no local load. The producer sells and buys energy in the Italian day-ahead market (MGP) and the three intraday auctions (MI1, MI2 and MI3). Later auctions can adjust the resulting position, while the battery has the double function of separating instantenous generation from future trades and of offering flexibility for real time operations when production differs from forecasts.

The core idea is to jointly optimize at each auction current market positions and expected future adjustments given available information and present (or expected) State of Charge (SoC). This allows to take advantage of price spreads while respecting physical constraints of our assets, speculating on market prices up until delivery. Only current auction trades are binding while future ones can still be updated as newer production forecasts are available. The actual battery state links the decisions across delivery intervals and days.

We formulate the problem as a Mixed-integer Linear Program (MILP) and solve it to optimality via GUROBI. The model maximizes expected remaining auction revenue less battery wear and trading costs, subject to energy conservation, exclusive (binary) charging/discharging, stored-energy bounds and the shared grid limit. The full optimization is in Appendix A.

We assess profits together with imbalance, curtailment, wear and ending stored energy.

2 The available data

2.1 Four farms and their historical records

The dataset covers four anonymous farms in the northern-Italian bidding zone. The supplied metadata describe fixed arrays and single-axis trackers and the battery are hypothetical study assets:

Farm	AC capacity (MW)	Declared technology	Battery (MWh)	Power (MW)
1	0.95328	Fixed array	0.75	0.5
345	1.52460	Fixed array	1.00	0.5
901	4.16000	Bifacial single-axis tracker	2.00	1.0
910	6.63800	Single-axis tracker	5.00	2.5

Their energy and power ratings, 7–93% state-of-charge range, 90% round-trip efficiency and connection limits are modeling assumptions. Exact farm coordinates are unavailable, so solar-position calculations use documented geographical proxies;

Historical records run from 1 January 2025 to 31 August 2026 on a 15-minute UTC grid: 58,368 timestamps for each farm. They contain AC power, historical *satellite* global horizontal irradiance (GHI), and historical diffuse fraction.

2.2 Updating deterministic weather forecasts

Given that positions must be traded mostly the day before delivery, obtaining accurate pv production forecasts for the next day is imperative. We use weather-to-pv models to obtain such estimates.

The main weather dataset comes from the ECMWF (European Centre for Medium-range Weather Forecast) archive containing 4 deterministic weather forecasts cycles per day (initialized every six hours) from 30 December 2024 to 31 August 2026. Each run simulates up until 48 hours from the initialization time. The weather fields include several irradiance components, temperature, wind and cloud cover.

Several cycles predict the same delivery interval with newly available information. This provides a record of forecast revision as delivery approaches. The four daily issues are successive information updates, each supplying one deterministic weather trajectory.

In the studies, forecasts are assumed to become available six or seven hours after initialization.

2.3 Weather ensembles

Two separate archives supply *50 weather members for the same issue and lead times*, with the same four farms and 0–48-hour lead coordinates:

These members represent alternative weather trajectories within one forecast issue. The spread supplies uncertainty information that a single deterministic forecast cannot describe. As more data becomes available, the ensemble can provide a more comprehensive view of potential outcomes, and act as training data for developing more sophisticated forecasting generation methods for stochastic methods.

2.4 Market observations

Historical clearing curves cover DA and the three intraday auctions, with imbalance-settlement prices used to value actual delivery deviations. Physical operation uses quarter-hours intervals resulting in 96 time steps per day, mirroring the products granularity.

The participation comparisons use actual future auction prices as a common price-information benchmark. This isolates choices about market access, production forecasts and physical flexibility under that benchmark. Their profits must be interpreted with this price-foresight assumption. Developing price forecasts is a further route toward operational decisions.

3 From weather information to production and decision paths

3.1 Selecting information at the auction cutoff

The study uses the following market timeline, in Italian local time. Decisions are taken five minutes before each gate:

Auction	Gate	Delivery covered	Latest typical weather issue
DA	Previous day, 12:00	Whole next day	Previous day, 00 UTC
IDA1	Previous day, 15:00	Whole next day	Previous day, 06 UTC
IDA2	Previous day, 22:00	Whole next day	Previous day, 12 UTC
IDA3	Delivery day, 10:00	Noon to midnight	Delivery day, 00 UTC

The last column follows the six-hour release assumption and requires sufficient remaining-horizon coverage. In our data weather forecasts have a 48-hour horizon allowing us to have prediction for trading in day ahead markets. At every cutoff, the deterministic forecast is updated to the latest eligible issue. Given the raw data size of storing all 4 cycles of the ensemble, only the 00 UTC is used for the ensemble studies.

3.2 Converting weather into AC generation

The baseline physical forecast is taken from the pvlib library. It first derives direct and diffuse irradiance components, then projects them onto the array or tracker plane. It estimates module temperature from air temperature, wind and plane-of-array irradiance, computes DC production and converts it to AC power. Output is bounded by the farm AC rating and forced to zero during the specified seasonal nighttime windows. Effective conversion parameters can be estimated from historical data.

Statistical and neural models provide another mapping from weather, calendar, lead-time and farm information to AC production. Historical power and irradiance support fitting and diagnosis; evaluation uses the forecast weather available for the delivery interval. Where learning uses historical irradiance, the shift to forecast irradiance is stated explicitly. Chronological training, selection and evaluation periods are kept distinct. Measured-only and reconstructed-target scores answer different questions.

Each ensemble weather member passes separately through the selected conversion model, producing a complete AC power path on the delivery grid. The study's 51-path input combines these 50 paths with the updating deterministic forecast. This allows us to generate a set of possible coherent scenarios instead of using independent sampling. Future studies, when more data becomes available, will investigate ways to generate similar temporally correlated trajectories for a more statistically sound stochastic optimization.

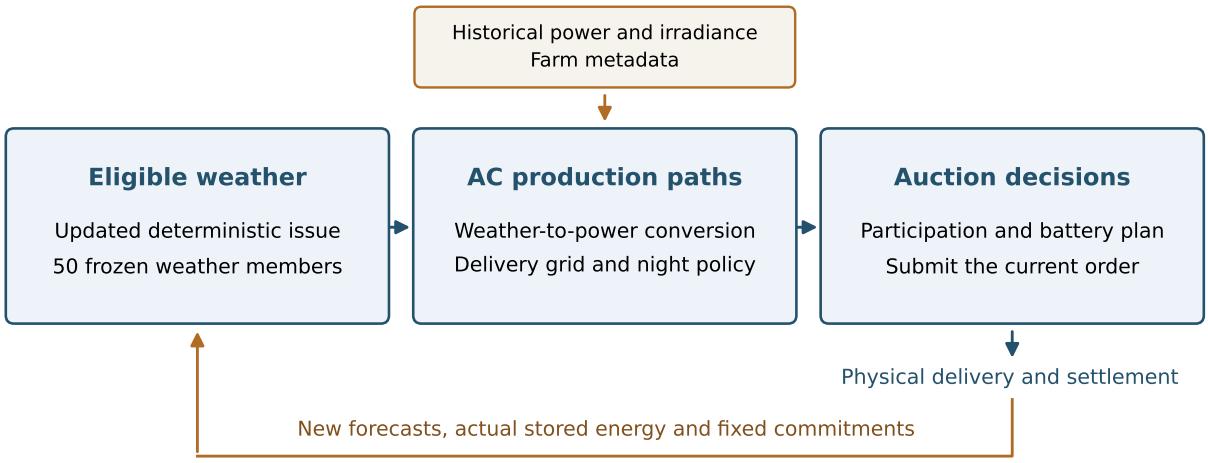


Figure 1: Historical records support the production mapping and evaluation. Eligible weather issues supply forecast paths; those paths support auction and physical decisions. Realized delivery supplies the carried battery state.

3.3 Turning forecast paths into a trading policy

At a given cutoff, a production path is a possible external input. A decision path consists of auction quantities and feasible battery/solar actions along the represented information history. Scenarios with the same information must share the decisions made before they can be distinguished.

3.3.1 Deterministic and ensemble formulations

The deterministic case optimizes against one eligible AC production forecast. It chooses auction quantities, charging/discharging and curtailment so that the forecast grid exchange exactly

matches the cumulative contracted position in each delivery interval. Energy, power, grid and terminal constraints apply to this one trajectory. At a later auction the same problem is solved again with updated information and the battery state appropriate to that decision.

The ensemble case uses 50 production paths from the preceding delivery day's 00 UTC weather issue plus the updating deterministic path, each weighted equally at 1/51. Auction quantities and the entire planned battery schedule are shared across all represented paths. Their common initial energy and charging/discharging imply a common stored-energy trajectory. Solar curtailment can differ with available production. For contracted energy Q_t, common battery powers c_t, d_t and path-specific curtailment $u_{s,t}$, exact delivery requires

$$(V_{s,t} - u_{s,t} + d_t - c_t)\Delta_t = Q_t, \qquad 0 \le u_{s,t} \le V_{s,t}, \qquad \forall s, t. \tag{1}$$

Thus one submitted position and battery plan must be feasible under every represented production path. The full formulation and its information-sharing constraints are in Appendix A.

	Deterministic case	Strict ensemble case
Production input	One updating forecast	50 frozen weather paths and one updating forecast
Trade and battery plan	One schedule	One schedule shared across all paths
PV curtailment	For the single forecast	Adapts to production in each path
Planned delivery	Exact for one forecast	Exact for every represented path
Main effect	Feasibility for the forecast trajectory	Feasibility over the intersection of pathwise constraints

Under the common actual-price curves used here, exact delivery makes useful solar injection identical across paths. It is therefore limited by the lowest available production at each interval: $0 \le i_t \le \min_s V_{s,t}$, whereas the deterministic case has $0 \le i_t \le V_{0,t}$. This envelope is an exact simplification under the shared-control assumptions. Its successive minima can come from different weather members; it is a feasibility bound rather than one meteorological trajectory. Averaging the ensemble first would instead give a single-forecast problem and remove this pathwise protection.

The objective retains a scenario-weighted expectation, but weights do not soften constraints imposed on every member. With identical prices and common grid exchange, the represented cash and physical costs coincide across paths; the ensemble changes the feasible decisions through production uncertainty. This explains the conservatism of this strict formulation. Allowing planned deviations or future battery actions conditional on new observations defines a different stochastic formulation.

3.4 Reoptimizing the remaining decisions

Integrated planning optimizes the remaining participating auctions together. Only the closing auction's quantity is submitted. At the next gate, that order is fixed and provisional later orders are reoptimized with new eligible forecasts. During delivery, commitment tracking adjusts physical battery power within its energy, power and grid limits. Actual SOC, submitted commitments, wear and transaction-fee state carry into subsequent decisions and days. A forecast projection supplies the starting state for tomorrow's plan while today's delivery is still in progress; the realized state governs execution.

4 Imbalances and real-time control

Even an exactly balanced forecast plan can produce a realized imbalance: actual exported energy differs from the submitted position, and the signed deviation is valued at the applicable settlement price. Our baseline is a *greedy grid-tracking policy*. In each quarter-hour it adjusts battery power to bring grid exchange toward the contracted target, discharging for a shortfall and charging for a surplus within available power and SOC limits. It releases scheduled curtailment when useful and redirects otherwise-curtailed PV into the battery when charging power and SOC headroom permit, while preserving the grid-exchange target. Remaining excess PV is curtailed. This local correction uses interval-average PV feedback in the replay; it does not optimize the consequences for later intervals. The current *model predictive control (MPC)* study instead reoptimizes over the remaining delivery day every 15 minutes, using the current battery state, fixed commitments and eligible deterministic or ensemble PV forecasts. It trades expected imbalance volume or forecast settlement value against wear and terminal energy, executes only the first action and repeats. The comparison asks whether anticipating later shortfalls and preserving flexibility improves delivery and operating profit relative to greedy tracking. The physical controller submits no new auction orders; its changed battery state can affect subsequent auction decisions.

5 Participation studies: a compact illustration

The four auctions give 15 nonempty participation profiles. We compare their integrated planning with sequential all-auction planning, which optimizes only the current auction's trade, and a perfect-production reference. Each profile carries its own SOC, commitments, wear and fees chronologically. The January 2026 comparison starts at 90% SOC and uses a 50% planning terminal target. Figure 2 summarizes farm 1. Actual future auction prices are shared across paths, so integrated planning also benefits from anticipating cross-auction spreads.

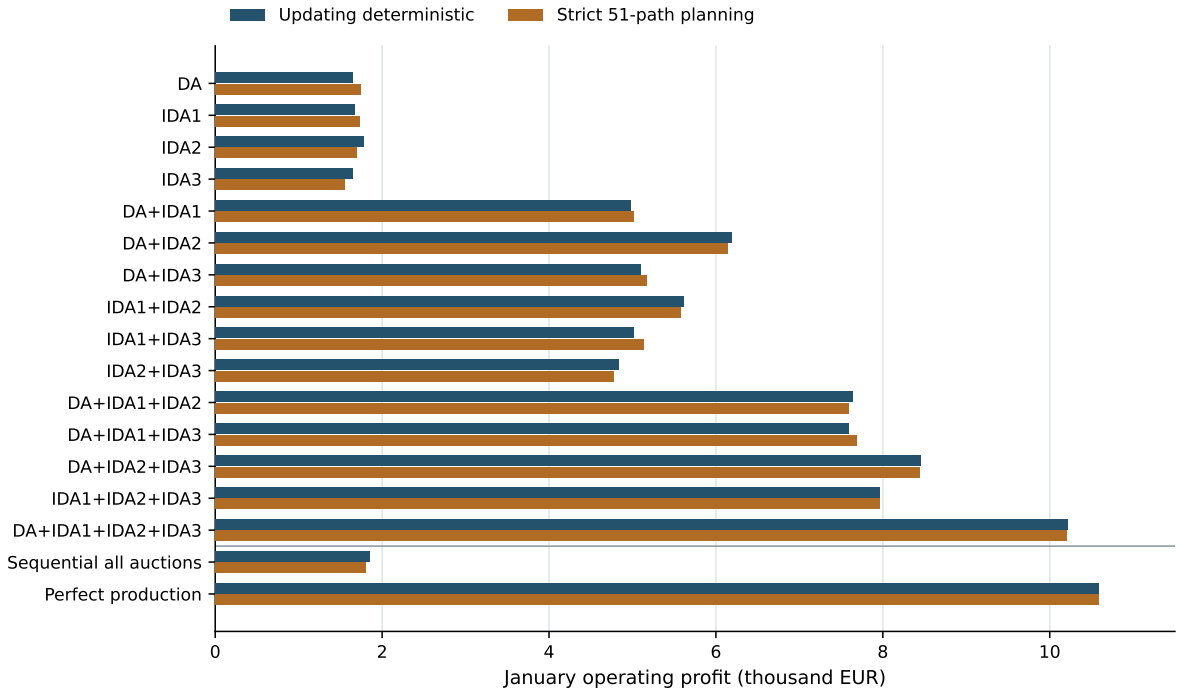


Figure 2: Farm 1, continuous January 2026: monthly operating profit for all 15 auction subsets and the two references. Paired bars compare the updating deterministic and strict 51-path formulations under actual-price foresight.

For farm 1's all-auction profile, ensemble planning reduces absolute imbalance from 5.818 to 1.577 MWh, while realized curtailment increases from zero to 0.775 MWh relative to deterministic planning. This illustrates the trade-off between tighter delivery tracking and curtailed solar energy. These curtailment totals are the residual after real-time feedback, which can redirect otherwise-curtailed PV into the battery when charging power and SOC headroom are available.

6 What this framework enables next

The available data connect forecast revision, production uncertainty and sequential participation within one physical chronology. The next questions are how forecast calibration and price prediction change those participation results, and how receding-horizon battery control uses new observations between auction gates. These developments build on the same data-to-path-to-decision structure. Longer comparisons must retain common dates, target provenance, information timing and ending-inventory accounting.

A Full residual auction optimization

A.1 Sets, parameters and signs

Let $t \in \mathcal{T}$ index the remaining physical intervals of a delivery day, Δ_t their duration in hours, $s \in \mathcal{S}$ scenarios with probabilities w_s, and $a \in \mathcal{A}_k$ auctions still open at decision k. $\mathcal{T}_a$ is auction a's eligible delivery set. Native products group physical intervals; their identifiers retain daylight-saving offsets.

Solar availability is $V_{s,t}$ MW AC, battery capacity E MWh, power limit $\bar{P}$ MW, energy bounds $E^{\min}, E^{\max}$, efficiencies η_c, η_d and hourly self-discharge σ. The grid limits are $P^{\text{imp}}, P^{\text{exp}}$. Q_t^{fix} is the signed MWh position from all earlier submitted quantities, including still-unpriced orders. Positive trade/grid signs mean sale/export. Auction price coefficients are $\pi_{a,s,t}$ in EUR/MWh.

Controls $c_{n,t}, d_{n,t}$ are charging/discharging MW; $z_{n,t}$ is a charging binary. Curtailment is $u_{m,t}$ MW, injection $i_{s,t}$ MW, grid exchange $g_{s,t}$ MW and internal energy $e_{s,t}$ MWh. $q_{a,n,t}$ is a signed auction adjustment in MWh. Node maps $n_t(s), n_a(s), m_t(s)$ encode observable history. Leaf values below refer to their node-owned variables, so indistinguishable histories cannot take different actions.

A.2 Complete base problem

For throughput wear $D_s = c_{\text{thr}} \sum_t (c_{s,t} + d_{s,t})\Delta_t$ and optional import cost $I_s = f \sum_t [-g_{s,t}]_+ \Delta_t$, the residual model is

$$\max \quad \sum_s w_s \left[\sum_{a \in \mathcal{A}_k} \sum_{t \in \mathcal{T}_a} \pi_{a,s,t} q_{a,n_a(s),t} - D_s - I_s - C_s^{\text{trade}} \right] \tag{2}$$

$$\text{s.t.} \quad i_{s,t} + u_{m_t(s),t} = V_{s,t}, \quad i_{s,t} \geq 0, \quad u_{m_t(s),t} \geq 0, \qquad \forall s,t, \tag{3}$$

$$g_{s,t} = i_{s,t} + d_{n_t(s),t} - c_{n_t(s),t}, \qquad \forall s,t, \tag{4}$$

$$-P^{\text{imp}} \leq g_{s,t} \leq P^{\text{exp}}, \qquad \forall s,t, \tag{5}$$

$$0 \leq c_{n,t} \leq \bar{P} z_{n,t}, \quad 0 \leq d_{n,t} \leq \bar{P}(1 - z_{n,t}), \quad z_{n,t} \in \{0,1\}, \qquad \forall n,t, \tag{6}$$

$$e_{s,t+1} = (1-\sigma)^{\Delta_t} e_{s,t} + \eta_c c_{n_t(s),t} \Delta_t - d_{n_t(s),t} \Delta_t / \eta_d, \qquad \forall s,t, \tag{7}$$

$$E^{\min} \leq e_{s,t} \leq E^{\max}, \quad e_{s,0} = e^k, \qquad \forall s,t, \tag{8}$$

$$Q_{s,t} = Q_t^{\text{fix}} + \sum_{a : t \in \mathcal{T}_a} q_{a,n_a(s),t}, \qquad \forall s,t, \tag{9}$$

$$g_{s,t} \Delta_t = Q_{s,t}, \qquad \forall s,t, \tag{10}$$

$$-P^{\text{imp}} \Delta_t \leq q_{a,n,t} \leq P^{\text{exp}} \Delta_t, \qquad \forall a,n,t \in \mathcal{T}_a, \tag{11}$$

$$-P^{\text{imp}} \Delta_t \leq Q_t^{\text{fix}} + \sum_{b \leq a : t \in \mathcal{T}_b} q_{b,n_b(s),t} \leq P^{\text{exp}} \Delta_t, \qquad \forall a,s,t, \tag{12}$$

$$q_{a,n,t}/\Delta_t = q_{a,n,t'}/\Delta_{t'}, \qquad t, t' \text{ in the same native product.} \tag{13}$$

Only eligible quantities exist. The last equality ties average power within a native product; on equal quarter-hours it ties MWh shares directly. Fixed initial energy can be observed or a forecast projection to the eligible boundary. A configured terminal equality $e_{s,T} = e_F$ and/or floor $e_{s,T} \geq e_{\text{floor}}$ is added where the study requires it. A planning terminal target does not itself force an algebraic feedback controller to end at that realized energy.

The month is the submission month, not necessarily the delivery month. Earlier auction cash and incurred wear are sunk and absent from (2). Only the closing auction's order is submitted; other quantities are revisable plans. Sequential planning replaces $\mathcal{A}_k$ by the single current auction.

B Battery wear, control and realized accounting

B.1 Wear models

For replacement value R, reference internal cycle depth d_0, life N and capacity E, AC throughput wear can be calibrated as

$$c_{\text{thr}} = \frac{R}{N d_0 E(1/\eta_c + \eta_d)}, \qquad D = c_{\text{thr}} \sum_t (c_t + d_t)\Delta_t. \tag{14}$$

An optional segmented depth model uses stress $\Phi(d) = \alpha d^\beta$, $\beta \geq 1$, J segments and marginal discharge cost

$$m_j = \frac{RJ}{\eta_d E}\{\Phi(j/J) - \Phi((j-1)/J)\}, \qquad D = \sum_{t,j} m_j d_{j,t} \Delta_t. \tag{15}$$

Segment energy transitions sum to aggregate energy; shallow-first charging and discharging priorities preserve cycle-depth memory. The segment vector carries between decisions and cannot generally be recovered from SOC alone. This is an approximation to cycle wear, not dynamic capacity fade or thermal/calendar aging [1].

B.2 Feedback, chronology and profit

Commitment tracking uses target $g_t^* = Q_t^{\text{sub}}/\Delta_t$ and interval PV. Write $\rho_t = (1-\sigma)^{\Delta_t}$ for energy retention. Feasible charge and discharge are bounded by

$$C_t = \min\left(\bar{P}, \frac{[E^{\max} - \rho_t e_t]_+}{\eta_c \Delta_t}\right), \quad D_t = \min\left(\bar{P}, \frac{\eta_d \left[\rho_t e_t - E^{\min}\right]_+}{\Delta_t}\right). \tag{16}$$

It adjusts net battery power toward the grid target, recovers curtailed PV where headroom permits, and enforces connection limits. Interval-average PV feedback is an idealized information assumption. Executed battery state, curtailment and deviations are retained at every interval.

Realized accounting uses

$$\delta_t = g_t^{\text{real}} \Delta_t - Q_t^{\text{sub}}, \tag{17}$$

$$J^{\text{real}} = \sum_{a,t} \pi_{a,t}^{\text{real}} q_{a,t}^{\text{sub}} + \sum_t \left(\lambda_t^+ [\delta_t]_+ - \lambda_t^- [-\delta_t]_+\right) - D^{\text{real}} - I^{\text{real}} - C^{\text{trade,real}}. \tag{18}$$

With a single settlement price this reduces to $\lambda_t \delta_t$. Forecast penalties, tail-risk terms and terminal continuation values are absent from realized cash. Final energy is reported separately. The state and binding positions carry across days; no daily physical reset occurs in a continuous comparison.

C Forecast construction and information timing

C.1 Issue, availability and delivery

For forecast initialization τ_v, lead h and availability A_v, define

$$t_{v,h} = \tau_v + h, \qquad A_v = \tau_v + \ell_{\text{release}}, \tag{19}$$

$$v^*(k) = \max\{v : A_v \leq k, \text{ required delivery horizon covered}\}. \tag{20}$$

Here ℓ_{release} is the study's assumed six- or seven-hour release lag; the main-body auction table illustrates the six-hour case. The deterministic production forecast at cutoff k is $V_{0,t}^{(k)} =$

$f_\theta(X_{v^*(k),t}, M)$, where M denotes farm assumptions. For delivery day d, let v_d be the preceding day's 00 UTC ensemble issue. The path input is

$$V_{s,t}^{(k)} = f_\theta(X_{v_d,s,t}, M), \quad s = 1, \ldots, 50, \qquad w_s = \frac{1}{51}, \quad s = 0, \ldots, 50. \tag{21}$$

The deterministic path updates while the weather-ensemble issue is fixed for the day. Model availability and observation delays are additional eligibility conditions; retrospective fitted models retain their historical-use label.

C.2 Physical and statistical production mappings

For GHI, horizontal beam irradiance BHI and solar zenith θ_z, the physical conversion uses the symbolic relationships

$$G_{\mathrm{DHI}} = G_{\mathrm{GHI}} - G_{\mathrm{BHI}}, \qquad G_{\mathrm{DNI}} \simeq G_{\mathrm{BHI}}/\cos\theta_z, \tag{22}$$

$$G_{\mathrm{POA}} = G_{\mathrm{beam}} + G_{\mathrm{sky}} + G_{\mathrm{ground}}, \tag{23}$$

$$T_{\mathrm{mod}} = T_{\mathrm{air}} + G_{\mathrm{POA}}/(U_0 + U_1 v), \tag{24}$$

$$P_{\mathrm{DC}} = \max\left(0, \frac{P_{\mathrm{AC,rated}}}{\eta_{\mathrm{ref}}} \frac{G_{\mathrm{POA}}}{1000}[1 + \gamma(T_{\mathrm{mod}} - T_{\mathrm{ref}})]\right), \tag{25}$$

$$V_t = \mathbf{1}_{\{t \text{ outside the seasonal night window}\}} \operatorname{clip}\big(\mathcal{C}(P_{\mathrm{DC}}), 0, P_{\mathrm{AC,rated}}\big). \tag{26}$$

Low-sun handling and valid irradiance bounds are required for the DNI conversion. $\mathcal{C}$ represents the selected DC-to-AC/output-loss mapping. The reference coefficients are $U_0 = 25$, $U_1 = 6.84$, $\gamma = -0.004/^\circ\mathrm{C}$ and $\eta_{\mathrm{ref}} = 0.96$; fitted historical conversions retain their own parameters. Tracker position and any rear contribution use explicit geometry assumptions.

A statistical model instead learns parametrized (with parameters θ) mappings from inputs to the normalized target $y_t = P_t/P_{\mathrm{AC,rated}}$:

$$\hat{\theta} = \arg\min_\theta \frac{1}{|\mathcal{D}_{\mathrm{train}}|} \sum_{t \in \mathcal{D}_{train}} \rho\big(f_\theta(X)_t - y_t\big) \tag{27}$$

Squared or Huber loss fits point predictions; normalization and preprocessing are learned from training data. Calendar, weather and lead/issue features must be available at the cutoff. A sequence model may use the whole eligible issued weather sequence, including future forecast covariates. Future observed production is reserved for evaluation or an explicit perfect-production reference.

For ensemble values x_s, probabilities w_s and realized power y, marginal quality can be measured by

$$\mathrm{CRPS} = \sum_s w_s|x_s - y| - \frac{1}{2}\sum_{s,r} w_s w_r |x_s - x_r|. \tag{28}$$

RMSE, MAE, bias and coverage provide complementary diagnostics. These scores evaluate forecasts separately from realized profit, physical feasibility and serial dependence. A calibrated collection of marginals requires an explicit temporal coupling before it defines new production paths for storage.

References

[1] B. Xu et al., *Factoring the Cycle Aging Cost of Batteries Participating in Electricity Markets*, 2017. https://arxiv.org/abs/1707.04567.